

## A contribution to the biogeography and taxonomy of two Anatolian mountain brook newts, *Neurergus barani* and *N. strauchii* (Amphibia: Salamandridae) using ecological niche modeling

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Received: 29.07.2020

Accepted/Published Online: 10.12.2020

Final Version: 18.01.2021

**Abstract:** The Anatolian newt, *Neurergus strauchii*, is an endemic Anatolian species. Until recently, *N. strauchii* was represented by three subspecies. It has been discussed within a recent phylogenetic study in which the subspecies *N. s. barani* is recommended to be evaluated as a cryptic but distinct species. To address this subject, we aimed to discuss the niche differentiation between *N. barani* and *N. strauchii* using geographical and bioclimatic aspects. All georeferenced data of *N. barani* and *N. strauchii* were used to estimate the potential distributions of these species in the Anatolian Peninsula. To evaluate their ecological niche differentiation, point-based analysis and niche similarity tests were done. Ecological niche modeling outcomes demonstrated a significant niche differentiation between *N. barani* and *N. strauchii*. Moreover, since these species are distributed in the east and west of the Euphrates Basin, this river might be considered as a geographic barrier that can cause isolation for these species. Lastly, we demonstrated their potential distributions for future with several scenarios. Our findings strengthened the results of the recent phylogenetic study and indicated the necessity of handling “*barani*” taxa at the species level. Moreover, these results contribute, as a piece of evidence, to the biodiversity of Anatolia where another endemic species lives.

**Key words:** Niche differentiation, isothermality, *Neurergus barani*, *Neurergus strauchii*, Salamandridae, Anatolia

### 1. Introduction

The ecological niche is a fundamental phenomenon that has serious contributions to the speciation mechanism in species' evolutionary history (Van Valen, 1976; Shapiro et al., 2016). The individuals in the same population that separated over time have not only been geographically differentiated but also have had remarkable changes in their genetic structure and external morphology, which might have been occurred due to their dynamic interactions with habitats. On the other hand, this process may have resulted in sympatric or allopatric speciation (Butlin et al., 2008). Geographical isolation or different ecological requirements are important drivers that have a critical role in this aspect with remarkable contributions (Kurnaz and Hosseinian Yousefkhani, 2019).

Ecological niche modeling (ENM) is a valuable technique to understand more information about conservation, ecology, distribution, geography, and evolutionary biology of a species (Guisan and Zimmermann, 2000; Araújo and Guisan, 2006; Phillips

et al., 2006). The maximum entropy algorithm (MaxEnt) utilizes only georeferenced species occurrence data with several environmental layers (especially bioclimatic ones); hence it is a popular and more preferable method to predict a species possible distribution (Guisan and Thuiller, 2005; Phillips et al., 2006; Elith et al., 2011). This preferred approach has been widely practiced for demonstrating the effects of bioclimatic variables on the niche divergence among taxa by many researchers during the last two decades (Peterson et al., 1999; Wiens, 2004; Raxworthy et al., 2007; Barve et al., 2011; Hosseinian Yousefkhani et al., 2016a; Heidari, 2019). Moreover, Anatolian Peninsula and Near East Asia have been receiving serious interest in examining niche differentiations recently (Gül, 2013; Hosseinian Yousefkhani et al., 2016b; 2019; Kurnaz and Hosseinian Yousefkhani, 2019; Gül, 2019; Şahin et al., 2020).

Mountain brook newts (*Neurergus* Cope, 1862) were represented by five species around the world and distributed in Turkey, Iran, and Iraq in the Middle East

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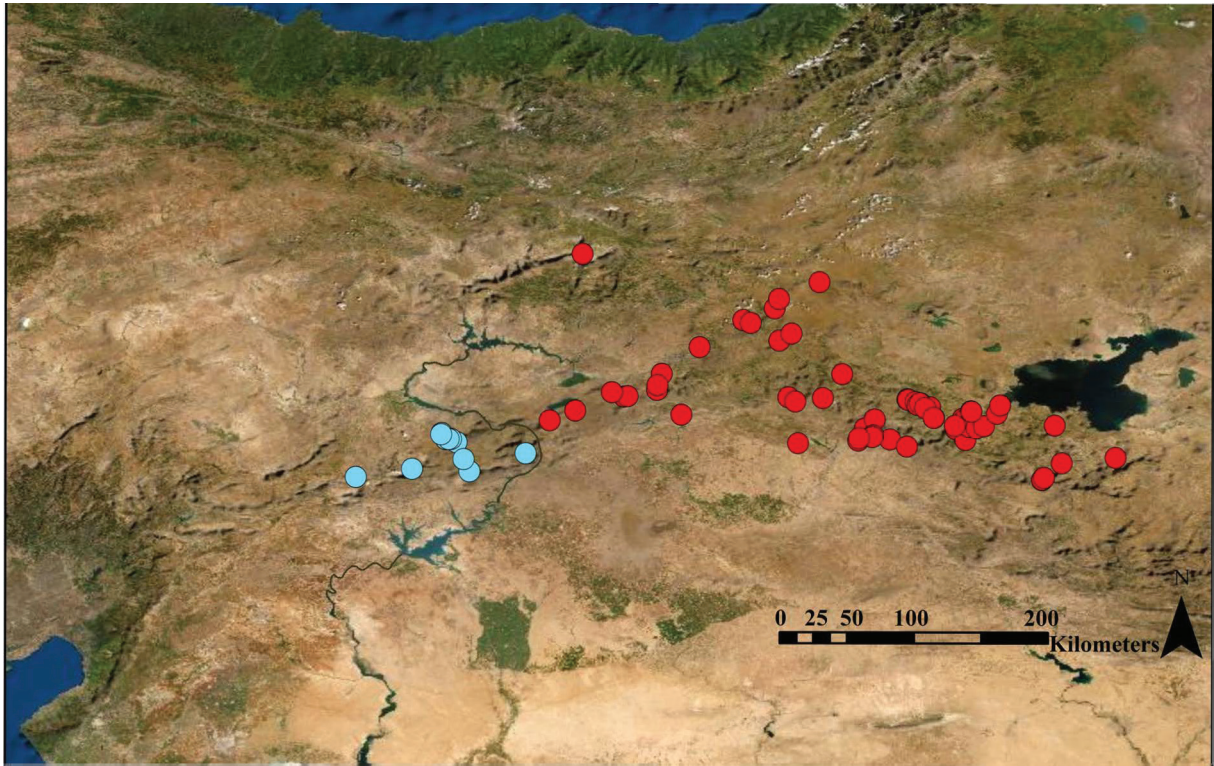
(Rancilhac et al., 2019). Based on previous studies, two species of *Neurergus* (*N. crocatus* and *N. strauchii*) have been reported from Anatolia (Özdemir et al., 2009, Kaya et al. 2012, Hendrix et al., 2014). Both *Neurergus* species are classified as VU (Vulnerable) according to the IUCN Red List of Threatened Species because of their limited distribution area and decreasing trends in their population densities (Papenfuss et al., 2009). *N. strauchii* was first described by Steindachner (1887) as *Molge strauchii* from the west of Lake Van (Muş). For years, *N. strauchii* had been considered as a subspecies of *N. crocatus* by different authors (Schmidt, 1939; Bodenheimer, 1944; Başoğlu and Özeti, 1973). However, Schmidtler and Schmidtler (1975), based on morphological analysis, proposed that *N. crocatus* and *N. strauchii* are two different species. These mentioned species were discussed in detail with the remarkable contributions from ENM approach recently and, as an output of this study, it was emphasized that ecological barriers play an important role in allopatric speciation process of these newts (Gül, 2019). With the new expeditions for *Neurergus* species, new taxonomic status for the discovered populations were suggested by researchers. For instance, the Kubbe Mountain population from the southeast of Malatya was recommended by Öz (1994) as the new subspecies *Neurergus strauchii barani*. The nominate subspecies, *N. s. strauchii* has a wider distribution area and spreads from the east of the Euphrates to the west and south of Lake Van (Schmidtler and Schmidtler, 1970; Baran and Öz, 1986; Öz, 1994; Pasmans et al., 2006; Schneider and Schneider, 2010; Çoşkun et al., 2013, Olgun et al., 2016; Yıldız et al., 2018), while *N. s. barani* is distributed in the west of the Euphrates River (Öz, 1994; Pasmans et al., 2006; Rancilhac et al., 2019). Along with these location record studies, the preliminary molecular-based phylogenetic studies revealed that there might be relatively significant genetic differentiation between these taxa (Steinfartz et al., 2002; Pasmans et al., 2006). After a decade, the first ecological niche modelling approach was conducted to *N. strauchii* (Tok et al., 2016). However, debates on the taxonomy of Anatolian *Neurergus* populations lasted for a significant time. Eventually, the recent phylogenetic study on this genus demonstrated that one of the subspecies of *N. strauchii* was promoted to be a distinct species, named *N. barani* (Rancilhac et al., 2019). The evaluation of *N. s. barani* to species rank (i.e. *N. barani*) has achieved due to *N. barani*'s significant genetic distance from *N. strauchii* with high-resolution sequencing data. However, *N. barani* lives in a very limited distribution area – an isolated group around Kubbe Mountain (West of the Euphrates), and there is no IUCN Red List assessment as *N. barani* has been considered a subspecies until 2019. Moreover, not only *N. strauchii* but also *N. barani* is an endemic species

for the Anatolian Peninsula, which hosts three exceptional biodiversity hotspots of the world (Mittermeier et al., 2004). This geographic region is in biodiversity crisis due to several reasons (i.e., habitat fragmentation, overgrazing, water use management, etc.) (Şekercioğlu et al., 2011), so evaluations of its uniqueness should stay on researchers' agendas.

Even there have been two studies predicting the potential distribution, ecological niche difference and future distribution of *Neurergus* species that have distribution in the Anatolian Peninsula (Tok et al., 2016; Gül, 2019), neither of them mentioned about *N. barani* populations as a unique species. For this reason, in the current study, we examined the *N. barani* taxon in terms of its potential spread, climate change, and ecological niche separation for the first time. The recent findings on these two species enable us to check if bioclimatic and geographic factors can influence the species delimitation. Therefore, our aims are i) to determine which bioclimatic variables are effective for the distribution of species, hence to predict highly suitable areas for distribution of *N. strauchii* and *N. barani*; ii) to examine niche divergence between these species; and iii) to give information about the possible future distribution of these two endemic species..

## 2. Materials and methods

This study was performed between 35°-45° eastern longitudes and 35°-41° northern latitudes including the southeastern and eastern parts of the Anatolian Peninsula, Turkey (Figure 1). A total of 74 occurrence data (11 for *N. barani* and 63 for *N. strauchii*) were gathered from the literature (Schmidtler and Schmidtler, 1970; Baran and Öz, 1986; Öz, 1994; Steinfartz et al., 2002; Arıkan et al., 2003; Bogaerts et al., 2006; Pasmans et al., 2006; Özdemir et al., 2009; Schneider and Schneider, 2010; Çoşkun et al., 2013; Olgun et al., 2015; Olgun et al., 2016; Akman et al., 2018; Yıldız et al., 2018; Rancilhac et al., 2019) (Appendix 1, Figure 1). We downloaded 19 bioclimatic variables in 1 km resolutions for current and future distributions (30 arc second) from WorldClim as v. 1.4 (Hijmans et al., 2005; available at [www.worldclim.org](http://www.worldclim.org)) (Appendix 2). The Community Climate System Model (CCSM4) was used for future predictions with its following scenarios. These are derived from greenhouse gas emission predictions named as representative concentration pathways (RCPs): RCP 4.5 and RCP 8.5. RCP 4.5 is used to describe global greenhouse gas emissions as long-term and short-lived. It explains the land use and land scenarios that stabilize the radiation force per square meter (approximately 650 ppm CO<sub>2</sub>-equivalent concentration) in 2100 (Thomson et al. 2011; Harris et al. 2014). On the other hand, RCP 8.5 represents the path to high greenhouse gas emissions



**Figure 1.** Location records for two species of *Neurergus* in the Anatolian Peninsula (blue points, *N. barani*, and red points, *N. strauchii*).

with a high radioactive forcing per square meter at the end of this century. For this scenario, the predicted probable temperature increase for the year 2100 is 2.6-4.8 °C (Riahi et al. 2011; Harris et al. 2014). Each bioclimatic parameter was masked to terrestrial zones of study area via Arc Toolbox embedded in ArcGIS ver. 10.3 (Esri, California, CA, USA). Pearson correlation between variables and the coordinates data patterns of the species were calculated in R-3.6.3 (R Core Team 2020) and highly correlated parameter pairs ( $r > |0.75|$ ) were excluded from analysis for eliminating the adverse consequences from other bioclimatic parameters (Appendix 3).

The potential climate suitability of the two species was modeled using MaxEnt 3.4.1 (Phillips et al., 2017) with the synthesis of occurrence records and reduced bioclimatic parameters (Elith et al., 2011). These reduced bioclimatic parameters are Bio 1 (Annual Mean Temperature), Bio 2 (Mean Diurnal Range), Bio 3 (isothermality), Bio 7 (temperature annual range), Bio 8 (mean temperature of wettest quarter), Bio 13 (precipitation of wettest month), and Bio 14 (precipitation of driest month). While studying the entire occurrence data, 75% of them were used for training and the remaining ones for test progress after removing the duplicated presence points. To construct the candidate models, the following settings used by Fathinia et al. (2020) were applied to MaxEnt, where the convergence threshold is 0.00001, maximum number of iterations is

500, the regularization multiplier is 1, maximum number of background points is 10000. Moreover, ten bootstrap replicates were run for studied species. To test the bioclimatic parameter importance, the jackknife test was applied in MaxEnt, which enables us to make a beneficial interpretation with the minimum presence records (Elith et al., 2006). Due to the recent advances in modeling process, we not only used the MaxEnt algorithm but also benefited from the NicheA 3.0 (Qiao et al., 2016) and ENMTools 1.3 (Warren et al., 2010) softwares for evaluating the candidate models; we selected the best model via Akaike Information Criterion corrected (AICc) for small sample sizes (Hurvich and Tsai, 1989). In addition to AICc, the power of the model was also determined by the values of the area under the receiver-operator (ROC) curve (AUC) (Raes and ter Steege, 2007; Gallien et al., 2012). According to Manel, Williams and Ormerod (2001), model scores are assessed as follows: AUC = 0.5 reflects a performance equivalent to random, AUC > 0.7 reflects a useful performance, AUC > 0.8 reflects a good performance and AUC ≥ 0.9 reflects an excellent performance. Finally, our model inputs were transformed into binary predictions using 10-percentile thresholding approach to visualize the “best” model (Perktaş et al., 2017).

Niche differentiation and similarity analyses implemented in ENMTools 1.3 software (Warren et al., 2010) and defined as identity tests and niche overlap were



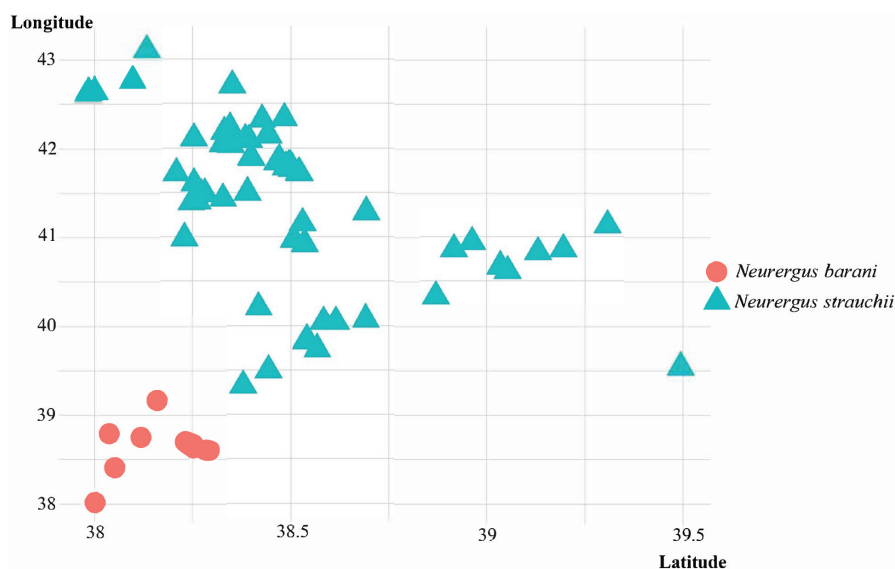
used to evaluate the differences between the potential climate suitability of both newt species. Two indices were measured for assessing climate identity: Hellinger's based  $I$  (Schoener and Gorman, 1968) and Schoener's  $D$  (Warren et al., 2008). The indices' scores vary between 0 (completely disjoint) and 1 (identical). To examine the significance of niche differentiation, 100 pseudo-replicates were generated with a pool of occurrence records of studied species. Lastly, the comparison was made from indices' scores of observed niche overlap ( $D_{H0}$  and  $I_{H0}$ ) and identity test ( $D_{H1}$  and  $I_{H1}$ ) (Gül, 2019; Şahin et al., 2020). The confidence level for  $D_{H1}$  and  $I_{H1}$  was 95% for this comparison.

### 3. Results

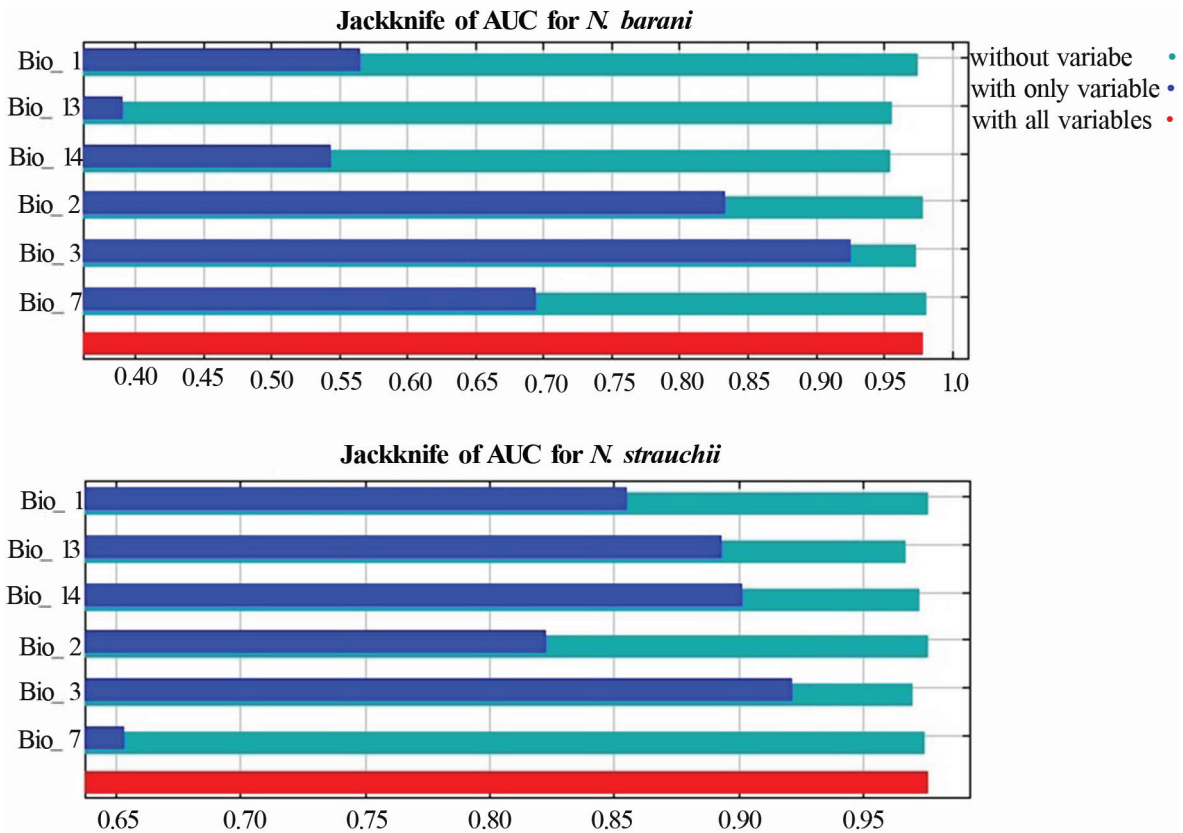
According to species distribution maps, it is clear that both species are separated geographically (Figure 1). Due to this fact, no overlapping occurrence data was seen between *N. barani* and *N. strauchii* (Figure 2). Modeling results obtained from AICc scores as decisive criteria for both species showed good distributional predictions (Figure 3). Moreover, for current distribution, models' AUC values were convincing for further analysis [ $0.951 \pm 0.01$  and  $0.968 \pm 0.018$  (mean  $\pm$  standard deviation) for *N. barani* and *N. strauchii*, respectively]. In addition, model AUC values were also highly good in both future scenarios, which represent moderate and extreme condition estimations for nature, respectively: rcp 4.5 [ $0.971 \pm 0.04$  and  $0.976 \pm 0.02$  (mean  $\pm$  standard deviation) for *N. barani* and *N. strauchii*, respectively) and rcp 8.5 ( $0.961 \pm 0.011$  and  $0.931 \pm 0.020$  (mean  $\pm$  standard deviation) for *N. barani* and *N. strauchii*, respectively)]. The climate suitability for *N. barani* was predicted to be in the west of the Euphrates

Basin (Figure 4a), while *N. strauchii* was predicted to be in the east of this basin, (Figure 4b), coinciding with the present occurrence of these two species. The future distribution of the two species is relatively narrower than under current bioclimatic conditions (Figure 5) in both scenarios. Towards the end of the century, the increase in carbon dioxide levels, radiation rates, and the greenhouse gases could possibly lead to a relatively small shrinkage of the potential distributional range of the two species. While these two species are compared under these future estimation scenarios, it might be speculated that the *N. barani* (Figure 5a, c) would be slightly less affected than *N. strauchii* (Figure 5b, d) from the effects of global warming. Therefore, it can be hypothesized that *N. strauchii* might be more sensible to increase in gas concentrations and air temperature.

The estimated distribution areas for studied *Neurergus* taxa are given in Figure 2. Despite the bioclimatic factors, which have a crucial role in the distribution of both newts are the same, bioclimatic envelopes are different: While Bio 3 (Isothermality) and Bio 14 (precipitation of driest month) are the most contributing variables, with 84.31% for *N. barani*, the variables Bio 3 and Bio 13 (precipitation of wettest month) are the most contributing ones, with 75.5% for *N. strauchii*. Although Bio 2 (Mean Diurnal Range) is a low correlated bioclimatic variable, it did not contribute to the potential distribution of both species (Table 1). In order to understand future distribution patterns of both species in two scenarios, the most contributing variables are given as follows: According to the results from RCP 4.5 scenario it is seen that while Bio 3 (Isothermality) and Bio 14 (precipitation of driest month) are the most contributing



**Figure 2.** The pattern of the coordinates data of both species, *N. barani* (red circle) and *N. strauchii* (blue triangle), with respect to latitude and longitude.



**Figure 3.** Relative predictive power of the six bioclimatic variables predicted by the jackknife of regularized training gain in MaxEnt model for both species (*Neurergus barani* and *N. strauchii*).

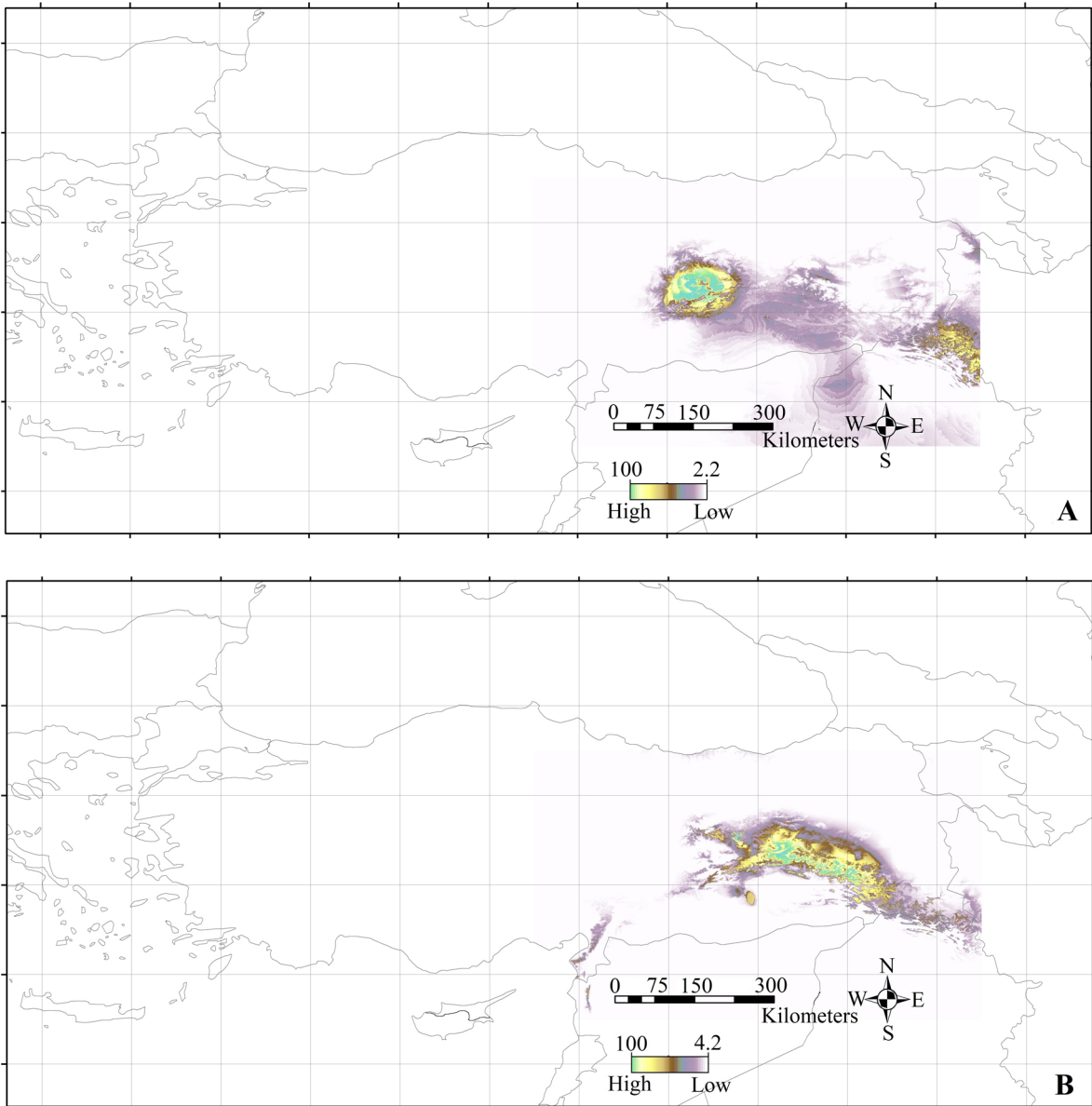
variables, with 81.6% for *N. barani*, the variables Bio 3 and Bio 13 (precipitation of wettest month) are the most contributing ones, with 76% for *N. strauchii*. On the other hand, based on the results of RCP 8.5 scenario, it is seen that while Bio 3 (isothermality) and Bio 14 (precipitation of driest month) are the most contributing variables, with 87.2% for *N. barani*, the variables Bio 3 and Bio 13 (Precipitation of Wettest Month) are the most contributing ones, with 75.7% for *N. strauchii*. Although Bio 2 (mean diurnal range) is a low correlated bioclimatic variable, it did not contribute to the potential distribution of both species for neither the current nor the future situations (Table 1). In addition to Bio 2 (mean diurnal range), Bio 8 also does not correlate with the other parameters; it was not included in the analysis due to odd spatial anomalies in the form of discontinuities between neighboring pixels in the absence of environmental gradients on the ground (Ashraf et al., 2017; Behroozian et al., 2020).

The outcomes of ecological niche divergence demonstrated that there is no evidence for niche overlap between these newt species (Schoener's  $D = 0.233$  and Hellinger's based  $I = 0.488$  for *N. barani* vs. *N. strauchii*). Moreover, the identity test denoted that our null hypothesis on niche overlap between *N. barani* and *N. strauchii*

should be rejected; hence, niche overlap between the two species was statistically different (t-test,  $df = 99$ ,  $P < 0.05$ ). The further evaluations showed that the predicted climate models for *N. barani* vs. *N. strauchii* were also significantly different; therefore, they were completely separated (Table 2) (Figure 6). According to these niche modeling results, it is clear that both species have been affected by not only different geography but also by different bioclimatic variables. However, due to the allopatric distribution of both species, the background test was not performed.

#### 4. Discussion

Species distribution is affected by several biotic and abiotic factors (Peterson, 2011). While these dynamics constitute their habitat requirements, they also provide a kind of adjustment that individuals of the species can live within this distribution area. Ecological niche modeling is used quite frequently to support taxonomic interpretations among close species, via potential distribution data (Nakazato et al., 2010; Kurnaz and Hosseinian Yousefkhani, 2019; Zhao et al., 2019). Niche distinction is crucial for cryptic species because each one invades a unique niche according to its ecological requirements (Zhao et al., 2019; Şahin et al., 2020). *N. barani* and *N. strauchii* are defined as

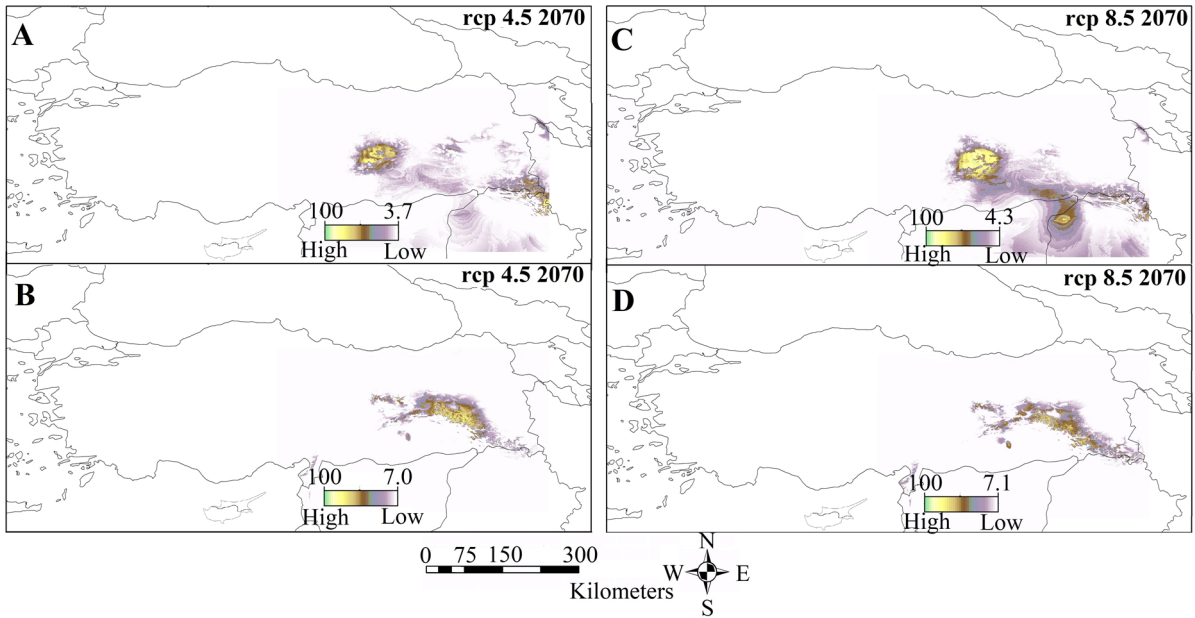


**Figure 4.** The range of current climate suitability predicted by MaxEnt model for A) *N. barani* and B) *N. strauchii* in the Anatolian Peninsula and Near East Asia.

cryptic species because of their phylogenetic position. This case has been supported with the latest phylogeography based study on this genus (Rancilhac et al., 2019). Here, a biogeographic evaluation was performed between *N. barani* and *N. strauchii* by assessing the climate difference dynamics in the current study that has not been previously done. According to our study, both point-based locality and potential distributional data demonstrated that there is no geographic overlap between *N. barani* and *N. strauchii*; both species have their unique distributional dimensions. Ecological niche differentiation, which is the most important factor in the speciation process, was

approved by niche identity analysis due to the momentous ecological niche alterations in these species.

The taxonomic assessment between *N. barani* and *N. strauchii* had been initially known at the subspecies level (Öz, 1994; Pasman et al., 2006; Özdemir et al., 2009; Hendrix et al., 2014). While some preliminary molecular-based studies conducted before showing differences phylogenetically between the *N. strauchii* and *N. barani* (Pasmans et al., 2006; Hendrix et al., 2014), Özdemir et al. (2009) claimed that there was no strong differentiation between these taxa based on phylogenetic interpretations from 12S and 16S rRNA gene fragments. However, none



**Figure 5.** The range of future climate suitability predicted with CCSM4 by MaxEnt for A,C) *N. barani* and B,D) *N. strauchii* in the Anatolian Peninsula and Near East Asia.

**Table 1.** Contribution of low correlated bioclimatic variables in species distribution modeling of *N. barani* and *N. strauchii*. Bioclimatic variables

|                                         | <i>N. barani</i> |                |            | <i>N. strauchii</i> |                |                |
|-----------------------------------------|------------------|----------------|------------|---------------------|----------------|----------------|
|                                         | Current          | Future rcp 4.5 | Future 8.5 | Current             | Future rcp 4.5 | Future rcp 8.5 |
| Bio_1 (Annual mean temperature)         | 1.2%             | 2.2%           | 0.6%       | 9.2%                | 8.1%           | 11.4%          |
| Bio_2 (Mean diurnal range)              | -                | -              | -          | -                   | -              | -              |
| Bio_3 (Isothermality)                   | 59.91%           | 57.4%          | 65.5%      | 39.8%               | 37.1%          | 44%            |
| Bio_7 (Temperature annual range)        | 3.8%             | 5.3%           | 2.6%       | 1.3%                | 2.1%           | 1.4%           |
| Bio_13 (Precipitation of wettest month) | 11.6%            | 10.9%          | 9.6%       | 35.7%               | 38.9%          | 31.7%          |
| Bio_14 (Precipitation of driest month)  | 24.4%            | 24.2%          | 21.7%      | 13.97%              | 13.8%          | 11.6%          |

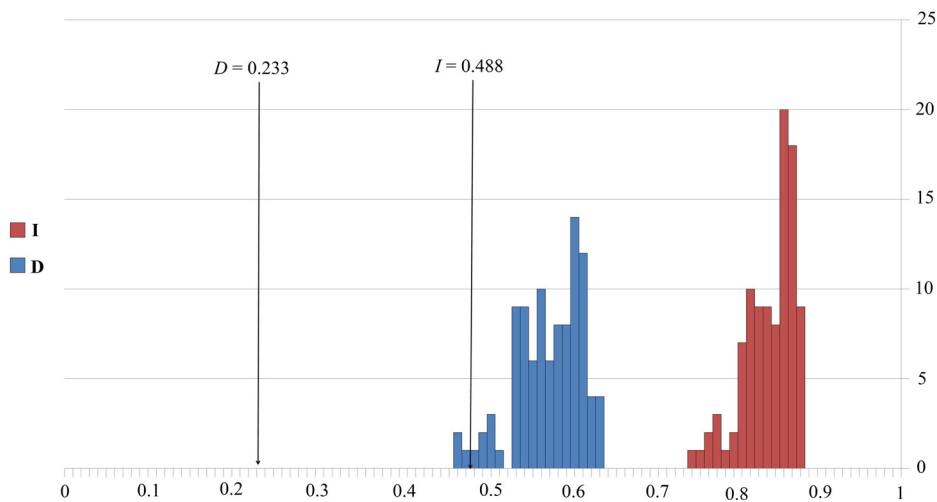
of these initial studies made a splitting recommendation at the species level between *N. barani* and *N. strauchii*. Lastly, a recent phylogenetic study recommended that these two subspecies be elevated to species rank because of their genetic distinction (Rancilhac et al., 2019). The results of our study is consistent with and they support the findings of this recent phylogenetic study, in a way that the suitable climate of these two taxa differs from each other, strengthening the validity of species rank of the two taxa. In line with these results, it can be thought that the Euphrates Basin, as a geographic barrier, caused both niche difference and a high genetic distance between these two taxa.

There has been an increasing interest in studying the interactions between a living organism and abiotic factors

in recent years. For instance, it is claimed that abiotic drivers like bioclimatic parameters might fortify speciation processes and adaptive divergence (Rissler et al., 2007). Our outcomes indicated that seven bioclimatic variables have an influence on climate differences and distributional limits between *N. barani* and *N. strauchii*. Up to now, Tok et al. (2016) provided the first explanation of habitat suitability for *Neurergus* species in Eastern Anatolia. Their study showed that eight different bioclimatic variables affect the spread of *N. strauchii*. Both Tok et al (2016) and we found that isothermality (Bio 3) made the highest contribution for *Neurergus* distribution in Turkey. Moreover, Gül (2019) pointed out that this bioclimatic variable (Bio3) is one of the climatologic influencers on the distribution and separation of *Neurergus* species. Due

**Table 2.** Two indices (Schoener's  $D$  and Hellinger's  $I$ ) based on niche overlap analysis between *N. barani* and *N. strauchii* in the Anatolian Peninsula.

| Comparisons                                     | Measured niche overlap |                            | Identity test<br>(significance threshold, $P < 0.05$ ) |                            |
|-------------------------------------------------|------------------------|----------------------------|--------------------------------------------------------|----------------------------|
|                                                 | Schoener's $D_{H0}$    | Hellinger's-based $I_{H0}$ | Schoener's $D_{HI}$                                    | Hellinger's-based $I_{HI}$ |
| <i>Neurergus sp. strauchii</i> vs <i>barani</i> | 0.233                  | 0.488                      | 0.569                                                  | 0.839                      |

**Figure 6.** Results of the identity tests ( $D$  and  $I$ ). The bars with different colors are calculated as the significance threshold of the replicates with identity test mode. Arrows refer to actual niche overlaps between *Neurergus barani* and *N. strauchii*.

to amphibians' life cycle, isothermality is expected to be a key bioclimatic variable, affecting amphibian distribution in several regions of the Earth (D'Amen et al., 2011; Hu et al., 2016).

Wellenreuther et al. (2012) suggested that three different endpoints exhibited the significance of niche differences between species. The first one is based on a calculation according to the niche similarities of studied species. Our results revealed that there is significant ecological differentiation between *N. barani* and *N. strauchii*, hence their niches differ from each other (Figure 4). The second endpoint is assessed via the identity test. Our identity test also showed that the ecological niches of these two species differ significantly (Figure 6). According to these findings, it might be claimed that there is a significant niche differentiation between these two *Neurergus* species. The third endpoint inferences with geography. The results in this study revealed an ecological niche distinction in an allopatric state (Figure 1 and Figure 2). Since the Euphrates River has acted as a geographic isolating barrier between the two taxa, a significant genetic distance has occurred between these two species during their evolutionary history. Consequently, this matter has led to the differentiation of the climatic preferences of these species.

Biogeography of *Neurergus* spp. in Turkey revealed that while the Cilo Mountains is a geographic barrier between *N. strauchii* and *N. crocatus* (Schmidtler and Schmidtler, 1970), Gül (2019) contributed to this outcome by examining the niche differentiation between these species. The Euphrates River made the same role between *N. barani* and *N. strauchii* providing a decisive effect in the separation of these taxa. It can be concluded that Euphrates River basins have a remarkable aspect to regulate microhabitat and vegetation dynamics along the river's flowing line for a geologically longer time (Guba and Glennie, 1998; Zaitchik et al., 2007). This current study's consequences are compatible with the taxonomic assessment of Rancilhac et al. (2019) in that two taxa could be evaluated at species level in terms of niche differentiation.

The future distribution projections of both *N. barani* and *N. strauchii* species display relative narrowings in their potential climate habitats (Figure 4 and Figure 5). This consequence was put forward due to the increase in carbon dioxide concentration levels, radiation rates and greenhouse gases; so that, the species will probably face remarkable habitat losses in their spread towards the end of this century. The relevant inference claiming that



habitat loss for *N. strauchii* could be very likely in the future was also stated by Tok et al. (2016).

As a result, it can be concluded that these two newt taxa are endemic to the Anatolian Peninsula. Compared to *N. strauchii*'s relatively wider distribution area, *N. barani* dominates a relatively narrower distribution. However, it is strongly recommended that these two taxa need to be protected. Although *N. strauchii* has been assessed as vulnerable (VU), *N. barani* has no conservation status yet. It is recommended that all populations of *N. barani* be evaluated and regained a conservation status. In other words, this current situation points out that these *Neurergus* populations may be at risk of extinction in the future unless the necessary measures are taken by governmental

authorities. Therefore, local and national stakeholders can establish conservation management programs compatible with climate forecasts and local habitat dynamics.

### Acknowledgments

We would like to thank MSc. Elnaz Najafimajd, who dedicated her academic life to *Neurergus* species, for her valuable contributions. Furthermore, we are also thankful to anonymous reviewers, for their valuable effort in the transition of this manuscript to a reliable study. In addition to them, we are also grateful to Ms. Crystal Day from Virginia Tech for her contribution as a native speaker regarding the redaction of this manuscript.

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**Appendix 1.** The occurrence data of two species retrieved from literature and used in the present study.

| Species                                           | Latitude | Longitude |
|---------------------------------------------------|----------|-----------|
| <i>Neurergus barani</i> , 38.291250, 38.595567    |          |           |
| <i>Neurergus barani</i> , 37.996656, 38.023486    |          |           |
| <i>Neurergus barani</i> , 38.230988, 38.694884    |          |           |
| <i>Neurergus barani</i> , 38.037347, 38.782209    |          |           |
| <i>Neurergus barani</i> , 38.117773, 38.742401    |          |           |
| <i>Neurergus barani</i> , 38.251100, 38.625733    |          |           |
| <i>Neurergus barani</i> , 38.251100, 38.659067    |          |           |
| <i>Neurergus barani</i> , 38.251100, 38.642400    |          |           |
| <i>Neurergus barani</i> , 38.284433, 38.592400    |          |           |
| <i>Neurergus barani</i> , 38.051467, 38.398812    |          |           |
| <i>Neurergus barani</i> , 38.158284, 39.158056    |          |           |
| <i>Neurergus strauchii</i> , 38.378718, 39.319843 |          |           |
| <i>Neurergus strauchii</i> , 38.443138, 39.490621 |          |           |
| <i>Neurergus strauchii</i> , 38.539766, 39.818201 |          |           |
| <i>Neurergus strauchii</i> , 38.541111, 39.841700 |          |           |
| <i>Neurergus strauchii</i> , 38.691783, 40.069539 |          |           |
| <i>Neurergus strauchii</i> , 38.583904, 40.038592 |          |           |
| <i>Neurergus strauchii</i> , 38.418577, 40.200562 |          |           |
| <i>Neurergus strauchii</i> , 38.870176, 40.323356 |          |           |
| <i>Neurergus strauchii</i> , 39.052875, 40.613620 |          |           |
| <i>Neurergus strauchii</i> , 39.034823, 40.664621 |          |           |
| <i>Neurergus strauchii</i> , 39.131168, 40.824087 |          |           |
| <i>Neurergus strauchii</i> , 38.916049, 40.856678 |          |           |
| <i>Neurergus strauchii</i> , 38.962977, 40.937868 |          |           |
| <i>Neurergus strauchii</i> , 38.228772, 40.982862 |          |           |
| <i>Neurergus strauchii</i> , 39.307546, 41.124622 |          |           |
| <i>Neurergus strauchii</i> , 38.327775, 41.429700 |          |           |
| <i>Neurergus strauchii</i> , 38.389472, 41.496469 |          |           |
| <i>Neurergus strauchii</i> , 38.254164, 41.598797 |          |           |
| <i>Neurergus strauchii</i> , 38.522689, 41.713703 |          |           |
| <i>Neurergus strauchii</i> , 38.208352, 41.710491 |          |           |
| <i>Neurergus strauchii</i> , 38.501801, 41.762878 |          |           |
| <i>Neurergus strauchii</i> , 38.494350, 41.779304 |          |           |
| <i>Neurergus strauchii</i> , 38.498188, 41.802687 |          |           |
| <i>Neurergus strauchii</i> , 38.470305, 41.865957 |          |           |
| <i>Neurergus strauchii</i> , 38.398755, 41.892270 |          |           |
| <i>Neurergus strauchii</i> , 38.394351, 42.086120 |          |           |

| Species                                           | Latitude | Longitude |
|---------------------------------------------------|----------|-----------|
| <i>Neurergus strauchii</i> , 38.442576, 42.142363 |          |           |
| <i>Neurergus strauchii</i> , 37.983379, 42.615573 |          |           |
| <i>Neurergus strauchii</i> , 38.000269, 42.629089 |          |           |
| <i>Neurergus strauchii</i> , 38.097481, 42.748431 |          |           |
| <i>Neurergus strauchii</i> , 38.567939, 39.734987 |          |           |
| <i>Neurergus strauchii</i> , 38.615772, 40.040365 |          |           |
| <i>Neurergus strauchii</i> , 39.195644, 40.854531 |          |           |
| <i>Neurergus strauchii</i> , 38.535613, 40.916265 |          |           |
| <i>Neurergus strauchii</i> , 38.529964, 41.147880 |          |           |
| <i>Neurergus strauchii</i> , 38.692886, 41.277219 |          |           |
| <i>Neurergus strauchii</i> , 38.281655, 41.486105 |          |           |
| <i>Neurergus strauchii</i> , 38.246899, 41.386618 |          |           |
| <i>Neurergus strauchii</i> , 38.384036, 42.095683 |          |           |
| <i>Neurergus strauchii</i> , 38.326026, 42.037550 |          |           |
| <i>Neurergus strauchii</i> , 38.252901, 42.104651 |          |           |
| <i>Neurergus strauchii</i> , 38.339466, 42.128234 |          |           |
| <i>Neurergus strauchii</i> , 38.330665, 42.177976 |          |           |
| <i>Neurergus strauchii</i> , 38.344796, 42.226470 |          |           |
| <i>Neurergus strauchii</i> , 38.443265, 42.141506 |          |           |
| <i>Neurergus strauchii</i> , 38.427679, 42.314025 |          |           |
| <i>Neurergus strauchii</i> , 38.350644, 42.701190 |          |           |
| <i>Neurergus strauchii</i> , 38.133685, 43.109870 |          |           |
| <i>Neurergus strauchii</i> , 39.494087, 39.538846 |          |           |
| <i>Neurergus strauchii</i> , 38.520514, 41.719666 |          |           |
| <i>Neurergus strauchii</i> , 38.503844, 41.769136 |          |           |
| <i>Neurergus strauchii</i> , 38.487552, 41.782472 |          |           |
| <i>Neurergus strauchii</i> , 38.496128, 41.799606 |          |           |
| <i>Neurergus strauchii</i> , 38.465377, 41.832778 |          |           |
| <i>Neurergus strauchii</i> , 38.399783, 41.889326 |          |           |
| <i>Neurergus strauchii</i> , 38.347902, 42.032711 |          |           |
| <i>Neurergus strauchii</i> , 37.986110, 42.615860 |          |           |
| <i>Neurergus strauchii</i> , 37.999337, 42.629760 |          |           |
| <i>Neurergus strauchii</i> , 38.483563, 42.334424 |          |           |
| <i>Neurergus strauchii</i> , 38.441582, 42.142876 |          |           |
| <i>Neurergus strauchii</i> , 38.507860, 40.965516 |          |           |
| <i>Neurergus strauchii</i> , 38.271631, 41.478799 |          |           |
| <i>Neurergus strauchii</i> , 38.263267, 41.391884 |          |           |



**Appendix 2.** Summary of environmental variables from the WorldClim data set and descriptions of the environmental variables.

| Abbreviation | Variable description                                       |
|--------------|------------------------------------------------------------|
| Bio 1        | Annual mean temperature                                    |
| Bio 2        | Mean diurnal range (mean of monthly (max temp - min temp)) |
| Bio 3        | Isothermality                                              |
| Bio 4        | Temperature seasonality                                    |
| Bio 5        | Max temperature of warmest month                           |
| Bio 6        | Min temperature of coldest month                           |
| Bio 7        | Temperature annual range                                   |
| Bio 8        | Mean temperature of wettest quarter                        |
| Bio 9        | Mean temperature of driest quarter                         |
| Bio 10       | Mean temperature of warmest quarter                        |
| Bio 11       | Mean temperature of coldest quarter                        |
| Bio 12       | Annual precipitation                                       |
| Bio 13       | Precipitation of wettest month                             |
| Bio 14       | Precipitation of driest month                              |
| Bio 15       | Precipitation seasonality (coefficient of variation)       |
| Bio 16       | Precipitation of wettest quarter                           |
| Bio 17       | Precipitation of driest quarter                            |
| Bio 18       | Precipitation of warmest quarter                           |
| Bio 19       | Precipitation of coldest quarter                           |

**Appendix 3.** Correlation matrix among bioclimatic variables used in the current study.

|       | bio1   | bio2   | bio3   | bio4   | bio5   | bio6   | bio7   | bio8   | bio9   | bio10  | bio11  | bio12  | bio13 | bio14  | bio15  | bio16  | bio17  | bio18 | bio19 |
|-------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|--------|-------|--------|--------|--------|--------|-------|-------|
| bio1  |        |        |        |        |        |        |        |        |        |        |        |        |       |        |        |        |        |       |       |
| bio2  | 0.488  |        |        |        |        |        |        |        |        |        |        |        |       |        |        |        |        |       |       |
| bio3  | 0.441  | 0.713  |        |        |        |        |        |        |        |        |        |        |       |        |        |        |        |       |       |
| bio4  | 0.154  | 0.127  | -0.567 |        |        |        |        |        |        |        |        |        |       |        |        |        |        |       |       |
| bio5  | 0.959  | 0.591  | 0.340  | 0.387  |        |        |        |        |        |        |        |        |       |        |        |        |        |       |       |
| bio6  | 0.975  | 0.388  | 0.499  | -0.037 | 0.881  |        |        |        |        |        |        |        |       |        |        |        |        |       |       |
| bio7  | 0.114  | 0.486  | -0.258 | 0.889  | 0.383  | -0.098 |        |        |        |        |        |        |       |        |        |        |        |       |       |
| bio8  | 0.308  | 0.290  | 0.438  | -0.259 | 0.225  | 0.315  | -0.140 |        |        |        |        |        |       |        |        |        |        |       |       |
| bio9  | 0.855  | 0.385  | 0.262  | 0.259  | 0.86   | 0.826  | 0.198  | 0.106  |        |        |        |        |       |        |        |        |        |       |       |
| bio10 | 0.981  | 0.484  | 0.309  | 0.336  | 0.987  | 0.923  | 0.274  | 0.233  | 0.867  |        |        |        |       |        |        |        |        |       |       |
| bio11 | 0.981  | 0.458  | 0.544  | -0.031 | 0.896  | 0.995  | -0.057 | 0.345  | 0.819  | 0.930  |        |        |       |        |        |        |        |       |       |
| bio12 | 0.299  | -0.714 | -0.533 | -0.059 | -0.358 | -0.225 | -0.313 | -0.383 | -0.224 | -0.291 | -0.276 |        |       |        |        |        |        |       |       |
| bio13 | -0.026 | 0.522  | -0.377 | 0.010  | -0.089 | 0.027  | -0.242 | -0.303 | -0.022 | -0.018 | -0.015 | 0.899  |       |        |        |        |        |       |       |
| bio14 | -0.677 | -0.452 | 0.209  | 0.384  | 0.744  | 0.613  | -0.368 | 0.098  | -0.747 | -0.717 | -0.610 | 0.293  | 0.044 |        |        |        |        |       |       |
| bio15 | 0.861  | 0.472  | 0.274  | 0.364  | 0.887  | 0.802  | 0.300  | -0.018 | 0.772  | 0.892  | 0.807  | -0.208 | 0.134 | -0.815 |        |        |        |       |       |
| bio16 | 0.002  | -0.521 | -0.388 | 0.0339 | -0.057 | 0.054  | -0.227 | 0.319  | 0.016  | 0.013  | 0.010  | 0.910  | 0.991 | 0.014  | 0.157  |        |        |       |       |
| bio17 | -0.690 | -0.490 | -0.198 | -0.443 | -0.773 | -0.611 | 0.433  | 0.103  | 0.743  | -0.741 | -0.612 | 0.33   | 0.066 | 0.991  | -0.845 | 0.038  |        |       |       |
| bio18 | -0.694 | -0.407 | -0.133 | -0.443 | -0.770 | -0.628 | -0.394 | 0.167  | -0.837 | -0.750 | -0.622 | 0.247  | 0.015 | 0.916  | -0.820 | -0.018 | 0.924  |       |       |
| bio19 | 0.311  | -0.340 | -0.211 | 0.051  | 0.244  | 0.365  | -0.199 | 0.299  | 0.348  | 0.311  | 0.318  | 0.762  | 0.892 | -0.289 | 0.411  | 0.918  | -0.256 | 0.324 |       |